

Editing and Unlearning

DATA 443/643 · Week 10, note 10.2

Week 10

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Ideas from CS224N 2025 interpretability and ROME (locate, then write). Original slides; those sources are not copied.

Lecture plan

- 1 Causal locate, then a rank-one write
- 2 Three probes: edit, neighbor, paraphrase
- 3 A set is not a key
- 4 Retain is the utility cost

Key goal: flipping one prompt is not an isolated fact.

First time: editing versus unlearning

- What. Edit: rewrite one fact. Unlearn: lower a set of behaviors. They are not the same experiment.
- Why. Retraining from scratch is too expensive. Serving a wrong object is a product bug.
- How. Locate the MLP, then $W \leftarrow W + uv^\top$ so $W'k_* = v_*$. Unlearn: continue train + retain set.
- Tradeoffs. + Cheap write. – Neighbors leak; paraphrases miss. Retain accuracy is the cost of unlearning.

Locate, then write

- CS224N: if a component stores the fact, intervening on it should change the object.
- ROME: mid-layer MLP on the last subject token.
- MEMIT spreads the write across several layers.

Locate, then write. One key, not a new pretraining run.



One outer product

$$W' = W + uv^T$$

Choose u, v so $W' k_* = v_*$
and old keys barely move.

That is one fact, not a topic delete.

Toy $d = 4$: a rank-one write is
 $d + 4d = 20$ numbers, not a
full fine-tune.

CounterFact-style columns

- Edit 8/10 looks like success.
- Neighbor 2/10: Italy moved. The write leaked.
- Retain 9/10: unrelated facts mostly sat still.

A flipped prompt is not an isolated fact.

probe	success
edit (France→Berlin)	8/10
neighbor (Italy)	2/10
retain (unrelated)	9/10

A key versus a set

Edit

One association. Lab 10: `facts["paris"] = "capital of Germany"`. Neighbors must stay.

Unlearn

A corpus. Forget-QA 90%→20% is not success if retain-QA 88%→21%.

What to take from this lecture

Three probes

Edit, neighbor, paraphrase. One prompt is not a method.

Not legal delete

Verbatim drop can hide latent knowledge.

Lab 10

A dict overwrite is the cartoon, not ROME.